

ALEXANDRU ALBU

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Swiss residence: C permit

EDUCATION

Karlsruhe Institute of Technology
M.Sc. in Machine Learning & Telematics
Oct 2014 – Oct 2016
75/120 ECTS completed

Karlsruhe Institute of Technology
B.Sc.
Oct 2011 – Aug 2014
Awarded maximum grade
Thesis: Radio propagation estimation based on 3D subject pose and location

SKILLS

Programming & Data
Java, C/C++, Python, SQL, Javascript

Web & Mobile
Node.js, React 18
Android MVVM, iOS UIKit

AI Integration
Google Gemini, Claude, GitHub Copilot

LANGUAGES

Romanian (Native)
English (C1, IELTS 7.0)
German (C1, DSH-2)
Chinese (A1)
Spanish (A1)

AWARDS

- Google Core Tech Impact Award (H1 '24)
- 3+ Spot Bonuses
- 15+ Peer Bonuses
- Internal Prototyping Contest Winner

EXPERIENCE

Google Zürich, Switzerland
Senior Software Engineer Jan 2017 – Present

- **Privacy & Security (2021-2026):** Designed and developed a privacy-aware framework capable of processing petabytes of data at a terabytes-per-second scale while adhering to user privacy settings.
- Spearheaded a data lineage project using C++, Python, and AI agents to analyze trillions of datasets and millions of nodes, drastically improving data labeling for AI training.
- Built an end-to-end monitoring pipeline that aggregates and serves gigabytes-per-second of privacy data with seconds-fast query capabilities.
- Led a pilot integration linking internal user data storage directly to Google Takeout, increasing compliance and user trust.
- **Assistant Memory (2019-2021):** Co-developed a funded, startup-like initiative allowing users to save and search contextual phone data using Google Search technology.
- Managed a complex Java stack comprising tens of thousands of lines of code, ensuring seamless backend data enrichment and syncing.
- **Calendar Team (2017-2019):** Co-authored a unified, high-performance syncing protocol API handling millions of queries per second across all Calendar clients.
- Conducted 100+ technical interviews and actively participated in university and community outreach programs.

Canel SA Iasi, Romania
Software Developer Apr 2010 – Feb 2013

- Wrote over 6,000 lines of C code from scratch for a single-threaded traffic light controller featuring RS485 communication, local menus, and remote internet configurability.
- Built the frontend and PHP/MySQL backend for a stadium access control system deployed across multiple venues.

PRIOR EXPERIENCE

Google	Software Engineering Intern, Zürich, CH	2016
Google	Software Engineering Intern, Zürich, CH	2015
Google	Software Engineering Intern, Mountain View, CA	2014